

PARSA AHMADI

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Education

University of Waterloo

Expected Graduation: Apr 2029

Bachelor of Applied Science in Mechatronics Engineering | Minor: Artificial Intelligence

Waterloo, ON

- Coursework: Data Structures and Algorithms, Digital Logic | GPA: 3.9/4.0

Technical Skills

Languages: Python, TypeScript, JavaScript, C, C++, C#, Java, SQL, HTML/CSS

Developer Tools: Azure, Azure DevOps, Git, AWS, Claude Code, Docker, Snowflake, Databricks, Kubernetes, Postman, PostgreSQL, SQL Server, MySQL, Redis, Jupyter Notebook, Jira

Frameworks/Libraries: React, Next.js, Angular, Blazor, ASP.NET, .NET, Entity Framework, Node.js, Express, PyTorch, TensorFlow, scikit-learn, XGBoost, NumPy, Pandas, OpenCV, ROS2, PyQt

Experience

SPS Commerce

May 2026 – Aug 2026

Software Engineering Intern – Data Pipeline

Minneapolis, MN

- Built a data-model-as-code **CI/CD platform** with TypeScript, **Azure DevOps**, **Docker**, and **AWS** to replace manual UI-based Sigma data model changes and deployments, applying source-controlled model updates to Test and Production, validating each release, and publishing generated model and metadata artifacts to **S3** buckets
- Enabled **customer-facing data sharing** across **Snowflake** and **Databricks** by exposing analytics datasets through Snowflake Data Listings and Databricks Delta Sharing, provisioning tables and schemas for direct customer access
- Improved **Snowflake** cost efficiency and performance by optimizing SDW/ODS data flows, precomputing dimension joins, and introducing incremental item/location processing, reducing warehouse compute costs by over **\$10,000** and downstream data movement by **25%**
- Developed a cloud-native **.NET 8 Kubernetes CronJob** that automated daily foreign exchange rate ingestion, including data extraction, compressed CSV generation, **Amazon S3** publishing, and **Snowflake** bulk loading, creating a reliable and repeatable financial data pipeline

BTNX

Sep 2025 – Dec 2025

Software Engineering Intern

Toronto, ON

- Developed a **full-stack ERP SaaS platform** using **React**, **TypeScript**, and **.NET**, building **30+** end-to-end pages and designing the normalized SQL Server schema, serving **1,000+** active users
- Reduced load times by **30%** across key modules by optimizing **LINQ** queries in **C#**, implementing **async/await** validation, and improving **SQL** performance by introducing table indexes, stored procedures, and a Redis lookup cache
- Engineered scalable **RESTful APIs** with **ASP.NET** and **Entity Framework** to handle **5,000+** daily requests, reducing downtime by **20%** and implementing integrations with Microsoft Graph and Amazon SP-API
- Built an AI-powered conference tracking module with **Blazor** and **Python**, training OCR and NLP models using **PyTorch** to extract business card data and auto-populate attendee profiles with **96%** accuracy
- Developed **Python scripts** to train and deploy an **XGBoost** classification model for diagnostic strips with ONNX Runtime inference integrated into a **Blazor** interface, achieving **99%** strip reader accuracy

Neurosnap

May 2025 – Aug 2025

Software Engineering Intern

Toronto, ON

- Built backend services in **Python** using **Flask**, integrating **ML pipelines** for molecular docking and sequence analysis, and reducing processing errors by more than **25%**
- Designed **PostgreSQL** schemas and indexed queries for storing protein structures, variant predictions, and experiment results, improving query performance by nearly **50%**
- Integrated **RESTful APIs** with **Python** and **Flask** for NeuroFold model inference and containerized batch prediction pipelines with **Docker**, automating research and saving laboratories **\$5,000+** annually

Waterloo Aerial Robotics Group (WARG)

May 2025 – Dec 2025

Autonomy Team Member

Waterloo, ON

- Developed computer vision modules for landing pad detection and obstacle avoidance, increasing detection accuracy to **90%** by training ML models with **OpenCV**, **TensorFlow**, and deploying inference pipelines in **C++** within **ROS2**
- Implemented real-time telemetry features in the IMACS 2.0 ground station, reducing communication latency from **150 ms** to **90 ms** by optimizing asynchronous data handling using **Python**, **PyQt**, and **ROS2**